

Evan Lamb

437-436-9827 | elamb@uwaterloo.ca | evanlamb.ca | github.com/evanlambb

EDUCATION

University of Waterloo

Bachelors of Computer Science, GPA: 3.9

Waterloo, ON

Sept. 2024 – Aug. 2028

EXPERIENCE

Software Engineer

Shopify

Jan. 2027 – Apr. 2027

Toronto, ON

- Incoming Winter 2027 intern; will contribute to scalable infrastructure supporting millions of merchants.

Software Engineer

Manulife

Jan. 2026 – Apr. 2026

Waterloo, ON

- Architected a custom automated data pipeline using Python and MySQL to extract and synchronize Salesforce records into New Relic, delivering a centralized dashboard that enhanced system observability for the team.
- Engineered a Co-pilot Studio agent utilizing natural language processing to automate the generation of standardized Jira tickets, eliminating manual administrative overhead and accelerating issue resolution.
- Containerized a custom Python application using Docker and deployed it to the cloud via Kubernetes, orchestrating the build and release pipeline through GitHub Actions CI/CD workflows.
- Optimized business intelligence workflows by developing SQL queries within Salesforce Marketing Cloud, extracting critical performance metrics to drive data-informed decisions for senior business stakeholders.

Software Developer

Royal Bank of Canada (RBC)

May 2025 – Aug. 2025

Toronto, ON

- Developed a forecasting agent using Python that ingested and processed time-series data from 91 Kubernetes hosted services, enabling \$400K in annual savings for a single team and demonstrating POC.
- Accelerated cross-team adoption by developing a web UI and backend on-boarding workflow that streamlined integration of the forecasting agent, reducing setup time for new teams using Flask REST APIs.
- Achieved 5% MAPE by creating an “Iterative Calibration” method simulating LLM backpropagation with mini chain-of-thought runs, reducing forecast bias and improving prediction accuracy.
- Secured an initial and follow-up meeting with EVP of Core Banking & Payments Technology by presenting a PowerPoint of potential enterprise-wide benefits, leading to deeper evaluation of performance and cost savings.

PROJECTS

Connect 4 AI | *AlphaZero, Pytorch, Python, NumPy*

Aug. 2026

- Implemented an AlphaZero-style game-playing agent from scratch that learns purely through self-play with no human game data, integrating Monte Carlo Tree Search with a policy/value neural network.
- Engineered the full self-play training loop, game generation, a symmetry-augmented replay buffer, batched network updates, and versioned, resumable checkpointing with a NaN/inf loss guard to prevent poisoned checkpoints.
- Developed an evaluation harness (arena match runner, generation-vs-generation win rates, anchored Elo with CSV/JSON export) that quantitatively confirmed each agent beats its baselines.

ML YouTube Chrome Extension | *Python, Logistic Regression, TF-IDF*

Jul. 2026

- Engineered a Chrome extension that instantly filters non-educational YouTube videos from user feeds, ensuring fast, zero-latency inference, by deploying a lightweight logistic regression model entirely client-side.
- Overcame short-text classification constraints for video titles of 10 words or less, outperforming continuous vector embedding models like FastText, by implementing a highly discriminative TF-IDF matrix using 1-2 n-grams.
- Architected a scalable, version-controlled ML data pipeline, successfully managing and building a 14,000-video broad dataset via the YouTube API and YouTube DOM scraping.
- Designed a seamless, unobtrusive user interface that preserves YouTube’s native grid layout, by developing smart DOM placeholders, session analytics, and a dynamic model strictness slider.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL, Java, JavaScript, HTML/CSS

Frameworks: React, Flask, FastAPI, JUnit, Node.js

Developer Tools: Git Version Control, Kubernetes, Github Actions, Docker, Jenkins

Libraries: PyTorch, Pandas, NumPy, FastText, Matplotlib, LangGraph, LangSmith